Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I expect to use all types of for loops, dataclasses, files, functions and almost everything else we have learned at some point in class. I would like to improve on my creation and calling of files as I am working on this project.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect the creation of the math functions to calculate the limiting reactant, theoretical yield and % yield to be the easiest for me because for loops is what I have had to practice the most this semester and what I am best at. The hardest thing for me will be making a file with every element, its atomic weight and other information and then importing it and calling it in my code. I didn't spend much time on that so I might struggle.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I should work on the file of elements first because that will take the most time and the rest of the functions are hard to test if I don't have that data.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

I have a lot of other tests and assignments so the best way to get around that is to start the code early so I have plenty of time to finish. If I get stuck, I can go to office hours, ask the discord or talk to my professor. I have no questions.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

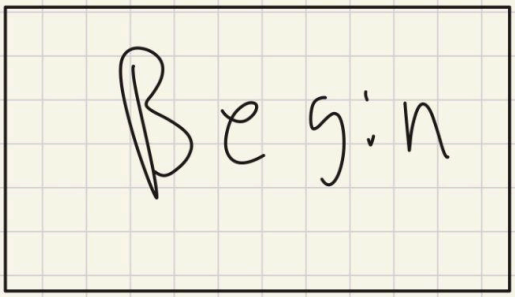
In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Welcome

(Home_page)

Ready to

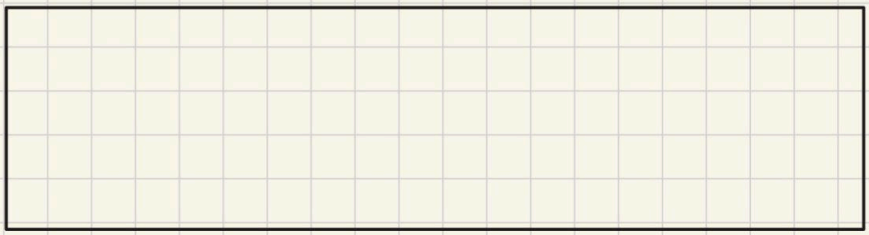


button takes you to next page

What is your balanced Chemical equation

(type_chem_equation)

Type here:



text box

back

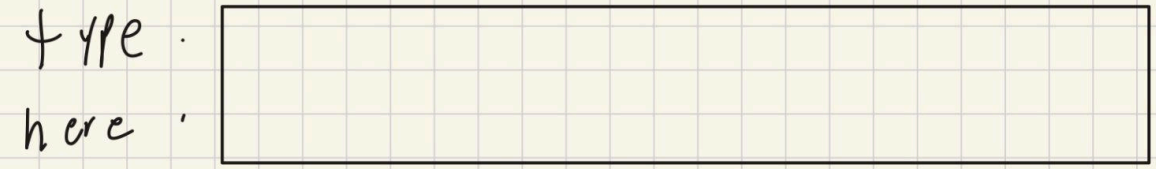
button takes you back to previous page

next

button takes you to next page

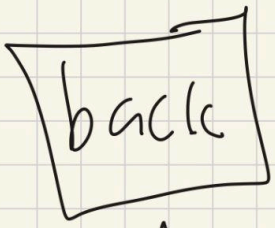
What are your reactant weights (type_weights)

type here:



(Your answer should be in a comma seperated list in order of your equation)

text box



button takes you back to previous page



button takes you to next page

Your limiting reactant is...

(find_limiting_reactant)

Compound / element

back

↑

button takes you back to previous page

Next

↑

button takes you to next page

Your theoretical yield is...

(find_theoretical_yield)

Theoretical Yield

back

↑

button takes you back to previous page

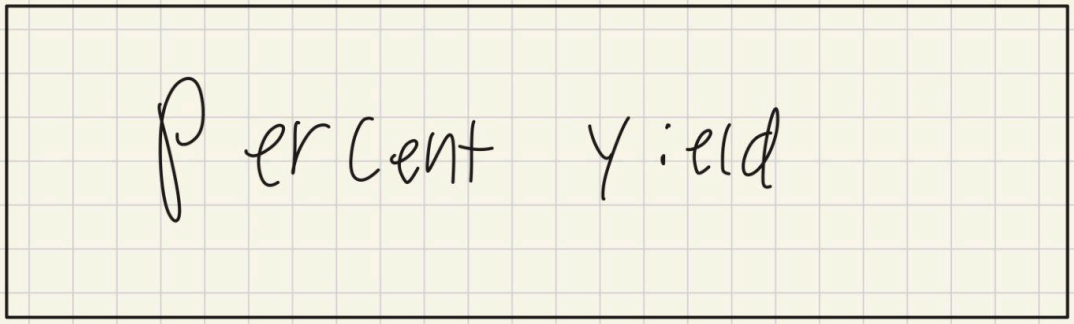
Next

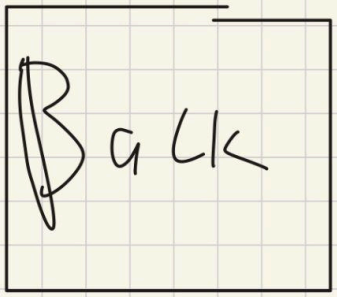
↑

button takes you to next page

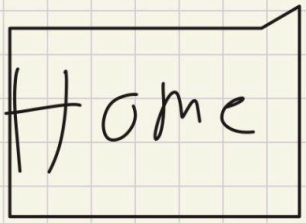
Your % yield is...

(find_percent_yield)





button takes you back to previous page



button takes you back to first page

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**This site will take in a balanced chemical equation and the masses of every reactant and product. It then will find the limiting reactant and calculate the theoretical yield and percent yield.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	attributes equation – String: Stores the balanced chemical equation entered by the user. chemicals – List of Chemical objects: Stores all reactants and products involved in the reaction. limiting_reactant – String: Stores the name of the reactant identified as the limiting reactant. It will be updated after calculations. theoretical_yield – Float: Stores the calculated maximum amount of product that can be produced. actual_yield – Float: Stores the actual amount of product entered by the user. percent_yield – Float: Stores the calculated percent yield.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified	Chemical Dataclass: name – String: Stores the chemical formula, such as H2 or O2. coefficient – Integer: Stores the coefficient of the

(if they are modified)	chemical in the balanced equation. molar_mass – Float: Stores the molar mass of the chemical in grams per mole. mass – Float: Stores the mass of the chemical entered by the user, measured in grams. is_reactant – Boolean: Identifies whether the chemical is a reactant or a product.
Routes (minimum 7) The names of your route functions	home – The homepage will welcome users and explain the purpose of the website. enter_equation – Allows the user to enter a balanced chemical equation. enter_reactants – Allows the user to enter the masses of each reactant. enter_products – Allows the user to enter the actual mass of the product produced. results – Displays the limiting reactant, theoretical yield, and percent yield. explanation – Explains how the program calculated its answers, including the steps involved. help – Provides instructions and examples to help users understand how to use the website.
Pages (minimum 5) A brief description of the pages in your site	Home Page: Welcomes the user and explains the purpose of the website. It will include a button to begin a new chemical reaction calculation. Chemical Equation Page: Allows the user to enter an already balanced chemical equation. Reactant Input Page: Allows the user to enter the mass of each reactant in grams. Product Input Page: Allows the user to enter the actual yield of the product in grams. Results Page: Displays the limiting reactant, theoretical yield, and percent yield.

User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	1. TextBox – Chemical Equation: Allows the user to enter a balanced chemical equation, such as 2H2 + O2 → 2H2O. 2. TextBox – Reactant Masses: Allows the user to enter the mass of each reactant in grams so the program can calculate the number of moles available. 3. TextBox – Actual Yield: Allows the user to enter the actual amount of product produced in grams so the program can calculate the percent yield. 4. Button – Calculate Results: When clicked, the program uses the information entered by the user to calculate the limiting reactant, theoretical yield, and percent yield. 5. Button – Clear Inputs: Allows the user to clear the information entered and start a new calculation.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. Limiting Reactant: An if statement will compare the mole ratios of each reactant. If one reactant has the smallest mole ratio, the program will identify it as the limiting reactant. 2. Valid Mass Input: An if statement will check whether the mass entered by the user is greater than zero. If the mass is zero or negative, the program will display an error message and ask the user to enter a valid number. 3. Percent Yield: An if statement will check whether the theoretical yield is greater than zero before calculating percent yield. If it is greater than zero, the program will calculate the percent yield. Otherwise, it will display an error message.
Loop (minimum 1) This loop must iterate over the	For Loop: The program will use a for loop to iterate through the list of Chemical objects stored in the

list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	Reaction dataclass. For each chemical, the loop will access its name, coefficient, molar mass, and mass. It will use this information to calculate the number of moles for each reactant and determine which reactant is limiting. This loop is important because it allows the program to process multiple chemicals without having to write separate calculations for each one.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

My project is described in the picture, give me ideas of what to fill out for the other boxes

This website will take in a balanced chemical equation and the masses of every reactant and product. It will then find the limiting reactant and calculate the theoretical yield and percent yield.

Some ideas for attributes are:

1. equation – String: Stores the balanced chemical equation entered by the user.
2. chemicals – List of Chemical objects: Stores all reactants and products involved in the reaction.
3. limiting_reactant – String: Stores the name of the reactant identified as the limiting reactant. It will be updated after calculations.
4. theoretical_yield – Float: Stores the calculated maximum amount of product that can be produced.
5. actual_yield – Float: Stores the actual amount of product entered by the user.
6. percent_yield – Float: Stores the calculated percent yield.

Some ideas for second dataclass

Chemical Dataclass:

1. name – String: Stores the chemical formula, such as H2 or O2.
2. coefficient – Integer: Stores the coefficient of the chemical in the balanced equation.
3. molar_mass – Float: Stores the molar mass of the chemical in grams per mole.
4. mass – Float: Stores the mass of the chemical entered by the user, measured in grams.
5. is_reactant – Boolean: Identifies whether the chemical is a reactant or a product.

The user will enter the chemical's formula and mass. The program will store its coefficient and molar mass and use those values to perform the necessary calculations.

Ideas for routes

1. home: The homepage will welcome users and explain the purpose of the website.
2. enter_equation: Allows the user to enter a balanced chemical equation.
3. enter_reactants: Allows the user to enter the masses of each reactant.
4. enter_products: Allows the user to enter the actual mass of the product produced.
5. results: Displays the limiting reactant, theoretical yield, and percent yield.
6. explanation: Explains how the program calculated its answers, including the steps involved.
7. help: Provides instructions and examples to help users understand how to use the website.

Ideas for pages

1. Home Page: Welcomes the user and explains the purpose of the website. It will include a button to begin a new chemical reaction calculation.
2. Chemical Equation Page: Allows the user to enter an already balanced chemical equation.
3. Reactant Input Page: Allows the user to enter the mass of each reactant in grams.
4. Product Input Page: Allows the user to enter the actual yield of the product in grams.
5. Results Page: Displays the limiting reactant, theoretical yield, and percent yield.
6. Explanation Page: Shows the steps and calculations used to determine the results.
7. Help Page: Provides instructions and a sample chemical reaction to help users

understand how to use the website.

Ideas for user inputs

1. TextBox – Chemical Equation: Allows the user to enter a balanced chemical equation, such as 2H2 + O2 → 2H2O.
2. TextBox – Reactant Masses: Allows the user to enter the mass of each reactant in grams so the program can calculate the number of moles available.
3. TextBox – Actual Yield: Allows the user to enter the actual amount of product produced in grams so the program can calculate the percent yield.
4. Button – Calculate Results: When clicked, the program uses the information entered by the user to calculate the limiting reactant, theoretical yield, and percent yield.
5. Button – Clear Inputs: Allows the user to clear the information entered and start a new calculation.

Ideas for if statements

The program will use if statements to check whether the reactant masses are valid, identify the limiting reactant by comparing mole ratios, and make sure the theoretical yield is greater than zero before calculating percent yield. If the mass is zero or negative, the program will display an error message. It will also identify the reactant with the smallest mole ratio as the limiting reactant.

Idea for a loop

The program will use a for loop to iterate through the list of Chemical objects in the Reaction dataclass. For each chemical, the loop will access its name, coefficient, molar mass, and mass. It will use this information to calculate the number of moles of each reactant and determine which reactant is limiting. This loop is important because it allows the program to process multiple chemicals without writing separate calculations for each one.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.